BUSINESS DATA ANALYSIS REPORT

Product Orders Dataset Analytics & SQL Solutions Framework

Prepared For: Senior Management / Analytics Lead	**Role:** Business / Data Analyst
Dataset: product_orders	**Scope:** Beginner to Advanced Analytical Queries

1. Overview & Executive Summary

This analytical report evaluates operational performance, customer purchase behaviors, product category dynamics, and revenue trends based on the product_orders transactional dataset. By applying structural SQL queries ranging from foundational data extraction to advanced analytical window functions, this report addresses critical business questions to support strategic decision-making.

key focal areas of this technical analysis include:

- **Order Fulfillment & Operational Health:** Monitoring cancelled orders and order volume distributions.
- **Customer Base Dynamics:** Identifying active customer engagement, top spending accounts, and cross-category purchasing behavioral patterns.
- **Product Performance & Category Analysis:** Evaluating pricing benchmarks, high-quantity purchase exceptions, and category filtering.
- **Regional & Temporal Revenue Metrics:** Grouping revenue metrics across geographic sales regions, tracking month-over-month (MoM) revenue growth, and calculating cumulative running totals.

1.1 Data Inspection Sample

Initial exploration query executed to understand the schema and row-level layout of the dataset:

```
SELECT * FROM product_orders LIMIT 5;
```

2. Technical SQL Query Analysis & Solutions

Level 1: Beginner Analytical Queries

Q1. Write a query to select all orders with OrderStatus = 'Cancelled'.

```
SELECT
    *
FROM
    product_orders
WHERE
    OrderStatus = 'Cancelled';
```

Business Insight & Analytical Notes: *Allows operational and customer service teams to audit cancelled transactions, analyze churn triggers, and measure order loss rates across fulfillment lines.*

Q2. Find the total number of distinct customers who placed an order.

```
SELECT
    COUNT(DISTINCT customerid) AS Total_Customers
FROM
    product_orders;
```

Business Insight & Analytical Notes: *Calculates total active unique buyers within the transactional period. Note: Modified from original input syntax to ensure valid aggregate calculation in standard SQL SQL engines.*

Q3. List all orders placed in the Electronics category, sorted by OrderDate descending.

```
SELECT
    *
FROM
    product_orders
WHERE
    category = 'Electronics'
ORDER BY orderdate DESC;
```

Business Insight & Analytical Notes: *Provides chronological visibility into the Electronics category, helping category managers track recent demand velocity and high-value item movements.*

Q4. Get the average UnitPrice of all products in the table.

```
SELECT
    ROUND(AVG(unitprice), 2) AS Average_UnitPrice
FROM
    product_orders;
```

Business Insight & Analytical Notes: *Establishes the baseline pricing benchmark across the entire product catalog, aiding in pricing strategy and margin assessment.*

Level 2: Intermediate Analytical Queries

Q5. Find the total revenue grouped by Region.

```
SELECT
    region,
    ROUND(SUM(totalprice), 2) AS Total_Revenue
FROM
    product_orders
GROUP BY region;
```

Business Insight & Analytical Notes: *Evaluates regional sales performance to assist regional managers in resource allocation, target setting, and regional logistics planning.*

Q6. Find the top 5 customers by total amount spent.

```
SELECT
    customerid,
    customername,
    ROUND(SUM(totalprice), 2) AS Total_amount_spent
FROM
    product_orders
GROUP BY customerid , customername
ORDER BY Total_amount_spent DESC
LIMIT 5;
```

Business Insight & Analytical Notes: *Identifies High-Value VIP Customers (Top Spend Groups) suitable for key account management, tailored loyalty programs, and retention initiatives.*

Q7. Write a query to find the number of orders per month (use OrderDate) for the year 2025.

```
SELECT
    EXTRACT(MONTH FROM OrderDate) AS order_month,
    COUNT(*) AS num_orders
FROM product_orders
WHERE EXTRACT(YEAR FROM OrderDate) = 2025
GROUP BY EXTRACT(MONTH FROM OrderDate)
ORDER BY order_month;
```

Business Insight & Analytical Notes: *Delivers seasonal order volume distribution across 2025, enabling operational forecasting for inventory procurement and warehouse staffing.*

Q8. Find all products that have never been ordered with Quantity greater than 5 (use a subquery or NOT IN).

```
SELECT DISTINCT
    productid, productname
FROM
    product_orders
WHERE
    productid NOT IN
    (SELECT DISTINCT
            productid
        FROM
            product_orders
        WHERE
            quantity > 5);
```

Business Insight & Analytical Notes: *Highlights retail items that fail to hit bulk purchases or high-quantity threshold orders (>5 units), pointing towards retail-focused or low-turnover items.*

Level 3: Advanced Analytical Queries

Q9. Write a query using a window function (RANK() or ROW_NUMBER()) to find the top 3 highest-value orders within each Region.

```
SELECT *
FROM (
    SELECT *,
           RANK() OVER (PARTITION BY Region ORDER BY TotalPrice DESC) AS rnk
```

```
    FROM product_orders
) ranked
WHERE rnk <= 3;
```

Business Insight & Analytical Notes: *Utilizes window partitioning to isolate outlier high-value transactions regionally, highlighting critical localized sales highlights without collapsing underlying transaction rows.*

Q10. Calculate the month-over-month growth in revenue (use LAG() to compare each month's total revenue to the previous month).

```
WITH monthly_revenue AS (
    SELECT
        DATE_FORMAT(OrderDate, '%Y-%m-01') AS order_month,
        ROUND(SUM(TotalPrice),2) AS revenue
    FROM product_orders
    GROUP BY DATE_FORMAT(OrderDate, '%Y-%m-01')
)
SELECT
    order_month,
    revenue,
    ROUND(LAG(revenue) OVER (ORDER BY order_month),2) AS prev_month_revenue,
    ROUND(revenue - LAG(revenue) OVER (ORDER BY order_month),2) AS revenue_change,
    ROUND(
        (revenue - LAG(revenue) OVER (ORDER BY order_month)) * 100.0
        / NULLIF(LAG(revenue) OVER (ORDER BY order_month), 0), 2
    ) AS pct_growth
FROM monthly_revenue
ORDER BY order_month;
```

Business Insight & Analytical Notes: *Calculates financial MoM revenue trajectories, revenue variance, and percentage growth rate using SQL windowing and defensive zero-division handing (NULLIF).*

Q11. Find customers who ordered from more than 3 different categories (use GROUP BY + HAVING COUNT(DISTINCT Category) > 3).

```
SELECT CustomerID, CustomerName, COUNT(DISTINCT Category) AS num_categories
FROM product_orders
GROUP BY CustomerID, CustomerName
HAVING COUNT(DISTINCT Category) > 3
ORDER BY num_categories DESC;
```

Business Insight & Analytical Notes: *Surfaces multi-category buyers with high engagement across product verticals, identifying core cross-selling opportunities.*

Q12. Write a query to compute a running (cumulative) total of revenue ordered by OrderDate using a window function.

```
SELECT
    OrderDate,
    TotalPrice,
    SUM(TotalPrice) OVER (ORDER BY OrderDate
                           ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS
```

```
running_total
FROM product_orders
ORDER BY OrderDate;
```

Business Insight & Analytical Notes: *Tracks progressive aggregate revenue build-up chronologically over time, essential for executive cash flow tracking and periodic revenue milestone benchmarking.*

3. Comprehensive Analysis Summary & Executive Takeaways

Based on the query structures and analytical techniques executed across the product_orders dataset, the following key insights and strategic business recommendations are synthesized:

- **Revenue & Growth Monitoring:** The combination of aggregate regional grouping and month-over-month (MoM) windowing (LAG) provides leadership with clear financial visibility. Tracking monthly growth percentages allows business leads to quickly react to revenue acceleration or slowdowns.
- **Customer Segmentation & Retention:** Evaluating top-spending customers (Top 5) and multi-category purchasers (>3 categories) isolates key revenue-generating client segments. Loyalty initiatives and personalized cross-selling campaigns should prioritize these groups.
- **Operational Efficiency & Risk Mitigation:** Filtering for cancelled orders enables supply chain and operations managers to perform root-cause analysis on order friction, product availability, or delivery bottlenecks.
- **Inventory & Pricing Optimization:** Analyzing bulk quantity thresholds (Quantity > 5) alongside average unit price (UnitPrice) benchmarking assists merchandising teams in adjusting volume discounts and optimizing product packaging for slow-moving stock.

Summary for Data / Business Analyst Practice:
This analytical suite covers foundational data extraction (WHERE, GROUP BY, HAVING), intermediate data transformation (EXTRACT, DATE_FORMAT, aggregation), and advanced data analytics (RANK, LAG, cumulative window functions). Implementing these standard analytical patterns ensures scalable, high-performance reporting for data-driven business operations.